

Question:

Sample Case study on class & objects Bank Account Management System: Problem Statement: Design a system to manage bank accounts. Classes: Account, Savings-Account, Current-Account. CO1 Attributes: Account number, balance, account type, interest rate (for savings account), overdraft limit (for current account), etc. Methods: Deposit, Withdraw, Calculate Interest, Transfer, etc. Object Usage: Create objects for each customer's account and perform transactions like deposits, withdrawals, and transfer.

Steps:

1. **Create** an Account class.
2. Declare accountNumber and balance as private variables.
3. Create a constructor to initialize the account number and balance.
4. Create getter and setter methods to access and modify account details.
5. Create methods for:
6. Deposit
7. Withdrawal
8. Transfer

9. Display account details
10. Create a SavingsAccount class that extends Account.
11. Add an interestRate variable and create a method to calculate interest.
12. Create a CurrentAccount class that extends Account.
13. Add an overdraftLimit variable.
14. Override the withdraw() method to allow withdrawal using the overdraft limit.
15. In the main() method, create one savings account and one current account.
16. Perform deposit and interest calculation on the savings account.
17. Perform withdrawal from the current account using the overdraft facility.
18. Transfer money from the savings account to the current account.
19. Display the final details of both accounts.
20. End the program.

Algorithm:

1. **Start**
2. Create a savings account with account number, balance, and interest rate.

3. Create a current account with account number, balance, and overdraft limit.
4. Display the savings account details.
5. Deposit the specified amount into the savings account.
6. Calculate interest using:
 $\text{Interest} = \text{Balance} \times \text{Interest Rate}$
7. Add the calculated interest to the savings account balance.
8. Display the current account details.
9. Check whether the withdrawal amount is within:
 $\text{Balance} + \text{Overdraft Limit}$
10. If valid, subtract the withdrawal amount from the current account balance; otherwise, reject the transaction.
11. Check whether the transfer amount is positive and available in the savings account.
12. If valid, subtract the amount from the savings account and add it to the current account.
13. Display the final balances of both accounts.
14. **Stop**

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Bank.java 1 X

C: > Users > venik > Documents > Records > week1 > J Bank.java > Account

1 package week1;

2

3 class Account {

4 private String accountNumber;

5 private double balance;

6

PROBLEMS 1OUTPUTDEBUG CONSOLETERMINALPORTS

PS C:\Users\venik> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMess

'week1.Bank'

Creating Accounts ---

Performing Savings Transactions ---

Account Number: SAV101 | Balance: \$10000.0

Deposited \$2000.0 successfully. New Balance: \$12000.0

Interest of \$600.0 added. New Balance: \$12600.0

Current Account Overdraft Test ---

Account Number: CUR202 | Balance: \$2000.0

Withdrew \$4500.0 (Current Account). Remaining Balance: \$-2500.0

Transfer Test ---

Transferred \$1500.0 to account CUR202.

Final Account States ---

Account Number: SAV101 | Balance: \$11100.0

Account Number: CUR202 | Balance: \$-1000.0

PS C:\Users\venik>

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